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📁 Current Position

2024 – Present 📌 **Postdoctoral Fellow**, Instituto de Ciencias Físicas (ICF), UNAM, Mexico.
Working on dark energy, cosmological tensions, and large-scale structure
Advisor: Dr. José Alberto Vázquez González

🔍 Research Interests

📌 **Cosmology:** Dark Energy, Large Scale Structure, Hubble tension, S_8 tension, Modified gravity, dark matter–dark energy interaction, scalar fields.

🎓 Education

- 2018 – 2024 📌 **Ph.D. in Physics**, Jamia Millia Islamia, Delhi, India
Thesis: *Cosmological non-standard field dynamics in light of observations*
Advisor: Dr. Somasri Sen
- 2014 – 2016 📌 **M.Sc. in Physics**, Jamia Millia Islamia, Delhi, India
CGPA: 8.04/10 (Distinction)
Thesis: *Poincaré–Cartan forms for extended phase space in curved background*
Supervisor: Prof. Pankaj Sharan
- 2011 – 2014 📌 **B.Sc. (Hons.) Physics**, Jamia Millia Islamia, Delhi, India
First Division (68%)

📖 Publications

Journal Articles

- 1 **S. A. Adil**, M. A. Zapata, Ö. Akarsu, and J. A. Vazquez, “Background-level reconstruction of scalar-field potentials from dark-energy histories and comparison with analytic potential families,” Mar. 2026.
arXiv: [2603.14693 \[astro-ph.CO\]](#).
- 2 M. A. Zapata, G. Garcia-Arroyo, **S. A. Adil**, and J. A. Vazquez, “How holographic is the dark energy? A spline nodal reconstruction approach,” *JCAP*, vol. 05, p. 058, 2026. 📄 DOI: [10.1088/1475-7516/2026/05/058](#). arXiv: [2507.22292 \[astro-ph.CO\]](#).
- 3 D. Kumar, A. Mitra, **S. A. Adil**, and A. A. Sen, “Exploring alternative cosmologies with the LSST: Simulated forecasts and current observational constraints,” *Phys. Rev. D*, vol. 111, no. 4, p. 043503, 2025.
📄 DOI: [10.1103/PhysRevD.111.043503](#). arXiv: [2406.06757 \[astro-ph.CO\]](#).
- 4 E. Specogna et al., “Updated Constraints on Omnipotent Dark Energy: A Comprehensive Analysis with CMB and BAO Data,” Apr. 2025. arXiv: [2504.17859 \[gr-qc\]](#).
- 5 **S. A. Adil**, M. G. Dainotti, and A. A. Sen, “Revisiting the concordance Λ CDM model using Gamma-Ray Bursts together with supernovae Ia and Planck data,” *JCAP*, vol. 08, p. 015, 2024. 📄 DOI: [10.1088/1475-7516/2024/08/015](#). arXiv: [2405.01452 \[astro-ph.HE\]](#).

- 6 **S. A. Adil** et al., “Omnipotent dark energy: A phenomenological answer to the Hubble tension,” *Phys. Rev. D*, vol. 109, no. 2, p. 023 527, 2024.  DOI: [10.1103/PhysRevD.109.023527](https://doi.org/10.1103/PhysRevD.109.023527). arXiv: [2306.08046](https://arxiv.org/abs/2306.08046) [[astro-ph.CO](#)].
- 7 N. Menci, **S. A. Adil**, U. Mukhopadhyay, A. A. Sen, and S. Vagnozzi, “Negative cosmological constant in the dark energy sector: tests from JWST photometric and spectroscopic observations of high-redshift galaxies,” *JCAP*, vol. 07, p. 072, 2024.  DOI: [10.1088/1475-7516/2024/07/072](https://doi.org/10.1088/1475-7516/2024/07/072). arXiv: [2401.12659](https://arxiv.org/abs/2401.12659) [[astro-ph.CO](#)].
- 8 **S. A. Adil**, U. Mukhopadhyay, A. A. Sen, and S. Vagnozzi, “Dark energy in light of the early JWST observations: case for a negative cosmological constant?” *JCAP*, vol. 10, p. 072, 2023.  DOI: [10.1088/1475-7516/2023/10/072](https://doi.org/10.1088/1475-7516/2023/10/072). arXiv: [2307.12763](https://arxiv.org/abs/2307.12763) [[astro-ph.CO](#)].
- 9 **S. A. Adil** et al., “S8 increases with effective redshift in Λ CDM cosmology,” *Mon. Not. Roy. Astron. Soc.*, vol. 528, no. 1, pp. L20–L26, 2023.  DOI: [10.1093/mnrasl/slad165](https://doi.org/10.1093/mnrasl/slad165). arXiv: [2303.06928](https://arxiv.org/abs/2303.06928) [[astro-ph.CO](#)].
- 10 A. Bassi, **S. A. Adil**, M. P. Rajvanshi, and A. A. Sen, “Cosmological evolution in bimetric gravity: observational constraints and LSS signatures,” *Eur. Phys. J. C*, vol. 83, no. 6, p. 525, 2023.  DOI: [10.1140/epjc/s10052-023-11707-4](https://doi.org/10.1140/epjc/s10052-023-11707-4). arXiv: [2301.11000](https://arxiv.org/abs/2301.11000) [[astro-ph.CO](#)].
- 11 Ruchika, **S. A. Adil**, K. Dutta, A. Mukherjee, and A. A. Sen, “Observational constraints on axion(s) dark energy with a cosmological constant,” *Phys. Dark Univ.*, vol. 40, p. 101 199, 2023.  DOI: [10.1016/j.dark.2023.101199](https://doi.org/10.1016/j.dark.2023.101199). arXiv: [2005.08813](https://arxiv.org/abs/2005.08813) [[astro-ph.CO](#)].
- 12 A. A. Sen, **S. A. Adil**, and S. Sen, “Do cosmological observations allow a negative Λ ?” *Mon. Not. Roy. Astron. Soc.*, vol. 518, no. 1, pp. 1098–1105, 2022.  DOI: [10.1093/mnras/stac2796](https://doi.org/10.1093/mnras/stac2796). arXiv: [2112.10641](https://arxiv.org/abs/2112.10641) [[astro-ph.CO](#)].
- 13 **S. A. Adil**, M. R. Gangopadhyay, M. Sami, and M. K. Sharma, “Late-time acceleration due to a generic modification of gravity and the Hubble tension,” *Phys. Rev. D*, vol. 104, no. 10, p. 103 534, 2021.  DOI: [10.1103/PhysRevD.104.103534](https://doi.org/10.1103/PhysRevD.104.103534). arXiv: [2106.03093](https://arxiv.org/abs/2106.03093) [[astro-ph.CO](#)].

Skills

Languages	 English (fluent), Spanish (poquito), Hindi, Urdu
Coding	 Python, Bash, $\LaTeX$
Scientific Tools	 CLASS, CAMB, MontePython, Cobaya, SIMPLEMC, GetDist
Inference Methods	 Emcee, MultiNest, PolyChord, George, GaPP
Computing	 Linux, Windows, HPC environments, MPI parallelization

Conferences

Nov 2024	 Presented a talk at <i>IX Essential Cosmology for the Next Generation</i> , Playa del Carmen, Mexico, on “Dark Energy in the Light of JWST Anomalies”.
	 Presented a colloquium talk at <i>Instituto de Ciencias Físicas (ICF), UNAM</i> , on “13.8 billion years in the making: the evolution of our Universe and its unsolved mysteries”.
Aug 2024	 Presented a talk at <i>21st Century Cosmology: Tensions, Anomalies & New Physics</i> , Ashoka University, India, on “Cosmological tensions and new physics in the dark energy sector”.
Dec 2022	 Presented a talk at <i>Indian Association for General Relativity and Gravitation (IAGRG)</i> , on “Is there an evidence of negative cosmological constant through recent cosmological observations?”.
Nov 2022	 Presented an e-poster at <i>Young Astronomers’ Meet</i> , ARIES, Nainital, titled “Evidence of an AdS vacua in the Universe”.

★ Conferences (continued)

- Oct 2022  Presented a talk at *Cosmology@CCSP_1*, on “Late-time acceleration due to a generic modification of gravity and Hubble tension”.
- Nov 2021  Presented a talk at *Midwest Relativity Meeting*, on “Late-time acceleration due to a generic modification of gravity and Hubble tension”.

Outreach & Mentorship

- 2026  Co-advisor of **Andoni Fontanet Fernández** for the BSc degree project titled, “**Simetrías del espacio-tiempo y transporte del momento angular en gases diluidos relativistas usando Teoría Cinética**”.
-  Co-advisor of **Andrea Valencia Reyes** for the BSc degree project titled, “**Algoritmos para el estudio de la energía oscura**”.
- 2024  Taught a cosmology program to school children ([link](#)).
- 2023  Co-mentored Dharmendra Kumar for Ph.D. course work.
- 2022  Co-mentored Amresh Verma for M.Sc. dissertation.
- 2021  Co-mentored Sonej Alam for Ph.D. course work.
-  Volunteered for science outreach program for **Square Kilometre Array (SKA)** in Vigyan Samagam.

Schools & Seminars

- Sept 2024  Regular participant in weekly seminars at **Instituto de Ciencias Físicas (ICF), UNAM**.
- July 2023  Attended *Summer School on Cosmology*, ICTP.
- July 2022  Attended *Summer School on Cosmology*, ICTP.
- Dec 2020  Attended the 31st meeting of **IAGRG**, IIT Gandhinagar.
- Sept 2020  Attended *Physics of the Early Universe*, ICTS (online).
- Nov 2019  Attended **SERB Preparatory School in Theoretical High Energy Physics (THEP)**, Tezpur University.

Teaching Experience

- Feb 2025  Teaching Assistant for the course “Machine Learning and Cosmology” (Instructor: Dr. José Alberto Vázquez González).
- Mar 2023  Taught physics to NEET/JEE aspirants at Orchid Institute, New Delhi ([link](#)).
- Aug 2021  Conducted B.Sc. Physics laboratory sessions, Department of Physics.

References

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